

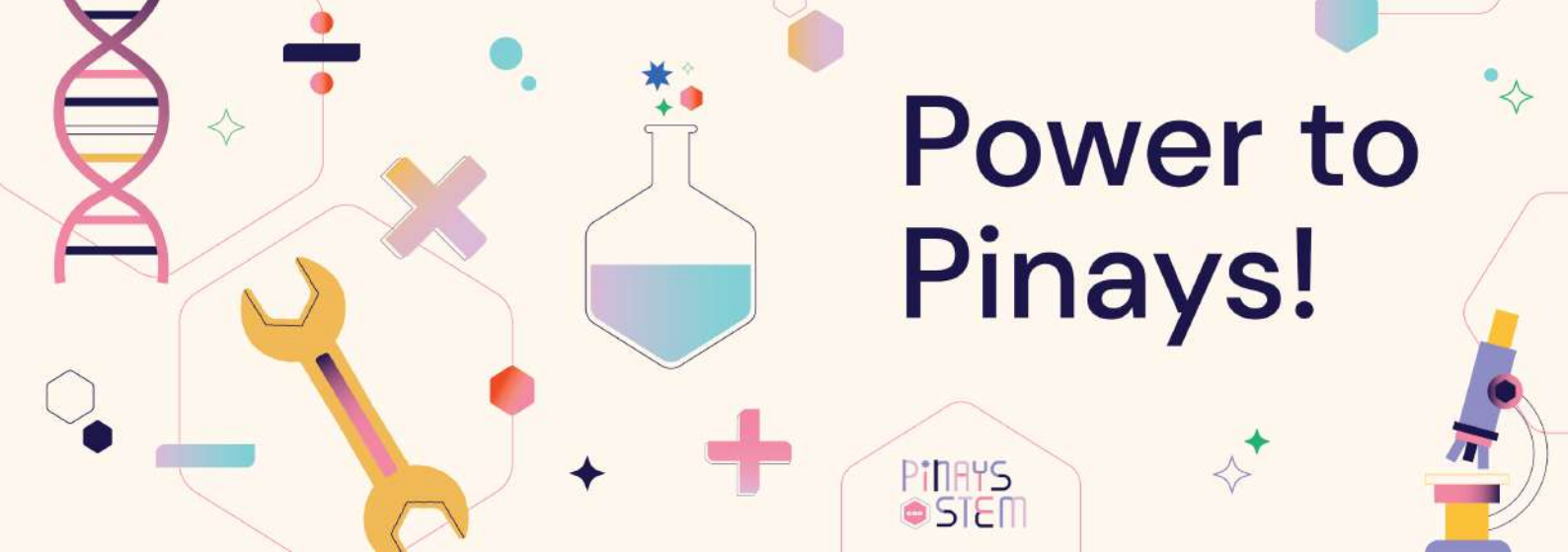


Toolkit Activity Guide for Teachers!



IN PARTNERSHIP WITH





Power to Pinays!

PINAYS
STEM

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NAME OF CONTACT PERSON

Unilab Foundation Inc.

EMAIL OF CONTACT PERSON

ISBN XXX (ebook)

Welcome, teachers!

Teachers play a crucial role in developing the next generation of scientists, designers, engineers, and mathematicians. By displaying positive beliefs and perceptions, they can influence interest and ensure success in the field. What teachers do in the classroom and the learning experiences they provide are critical components of encouraging students to pursue STEM careers.

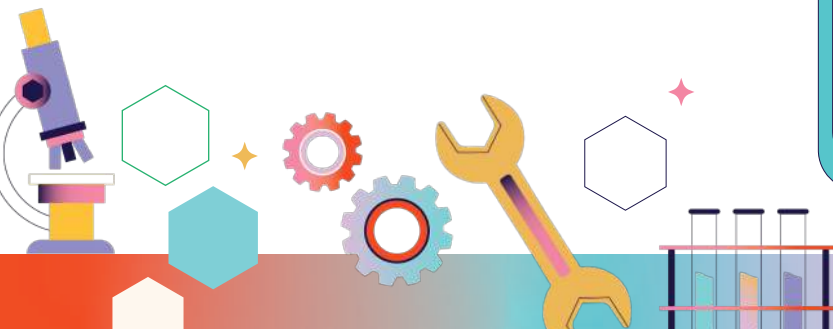
The Pinays Can STEM toolkit is a learning resource for Filipina students who want to take their STEM path to the next level! Use the weekly activities as a guide to help students on their STEM journey.

What is included in the Teacher's Kit?

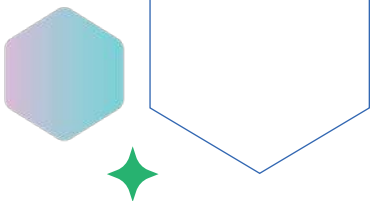
1. Activity guide (this document)
2. Activities form (for notes)
3. Pinays Can STEM Toolkit (Teacher's copy)
4. Slambook (Teacher's copy)
5. PowerPoint presentation template
6. Document containing links to:
7. Toolkit & Slambook, which will be shared with students
8. Tests (Pre-test, Posttest, Evaluation)

NOTE

1. You may also have 1-on-1 sessions with your students if you would like to get to know more about their STEM goals!
2. To help Unilab Foundation and Philippine Commission on Women with future research and policies, we hope you can note down and gather insights from the discussion through the activities form.



How to use this toolkit

- 
-  For easier navigation, the toolkit includes clickable buttons that allow you to skip from one page to another and redirect you to online resources.
 -  This activity guide is designed to help teachers use the toolkit more effectively. It should be noted that the toolkit and slambook are for students to keep and use beyond the activities listed.
 -  The activity guide is only for a one-month engagement, but teachers are free to create their own activities.
 -  Teachers may also prepare their own presentation for the sessions to accompany their toolkit discussion with students. The template can be found in your Teacher's Kit.
 -  We strongly encourage teachers to actively take notes and fill out our activity form so that we can compile all insights. Please send the completed form to Pinays Can STEM at pinayscanstem@gmail.com with the subject "[LAST NAME, FIRST NAME] SCHOOL – PCS Toolkit Activities."
 -  For each activity or discussion, guide questions will be provided, but teachers are free to add their own.
 -  Before beginning any toolkit-related activities, students will be given a pre-test. Following completion of the activities, a post-test will be administered to assess the student's confidence and interest. An evaluation form is included to determine the usefulness of the contents as well as what can be improved. The findings of these tests will assist the Unilab Foundation and the Philippines Commission on Women in improving policies, research, curriculum, and gender equality in STEM education.
 -  Please do not alter, copy, distribute, or claim ownership of the toolkit in any way. The material is solely the property of the Unilab Foundation and the Philippine Commission on Women. You are allowed to print this activity guide for easier facilitation.
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Repe the advocacy

Printables and online posters and templates

Join the fun online! Use these posters and become advocates of STEM yourselves!

DOWNLOAD



Cover photo



Profile picture frame



Pinays Can STEM poster

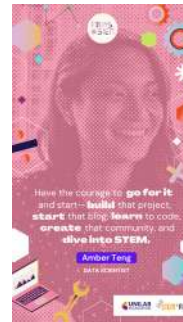


Printable poster

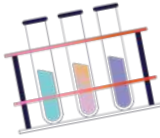
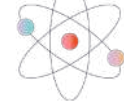
Desktop, phone, tablet wallpapers

Bring your inspiration wherever you go. Download these wallpapers for your phones, tablets, and computers!

DOWNLOAD



Articles to inspire your students



[Empowering Pinay Students at Home](#)

While online education can be overwhelming, teaching in a virtual classroom allows educational institutions to re-examine and innovate ways to transform the Philippine educational system. Here are some ideas for teaching STEM effectively even during the pandemic, as well as some inspiring words from our Pinays.



[I CAN: Bloom Where I Grow](#)

Every girl possesses the ability to transform “I cans” into “I have’s.” International Women’s Month may be over but we’re still here to lend some valuable ‘I CAN’ phrases to help us girls to not just bloom where we’re planted, but bloom where we belong.



[Our Sherones: The First Pinays of STEM](#)

Throughout history, women have long been cast in the shadows of their male counterparts—and the same can be said for our Filipina STEM pioneers. Even if we don’t know their names (yet), they have undoubtedly paved the way for women to enter spaces previously occupied by men.



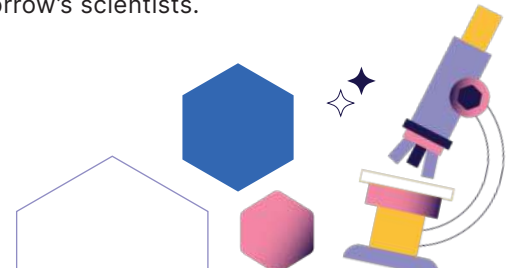
[Miss Represented: KickSTARTing gender equality in schools](#)

One of the most important ongoing goals for STEM girls stems from the UN Sustainable Development Goals (SDGs), which aim to achieve gender equality and women empowerment—a long-term plan that has yet to bear fruit for global girls in academia. Even though gender disparities are an ongoing issue in and out of the classroom, what we need is a cultural shift to get the ball rolling—a change that won’t happen overnight.



[Best in class: Preparing SHS students for STEM](#)

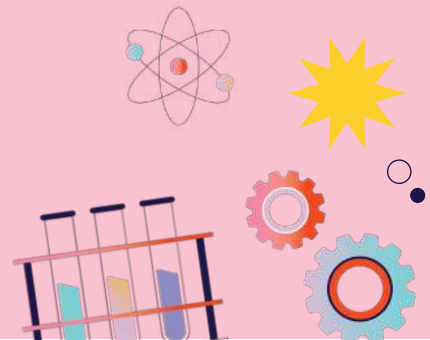
We teamed up with some study buddies from the non-profit organization Kababaihan Para Sa Siyénsiyá (@siyensiya.ph) to walk us through their personal STEM curriculum—and their testimonies for today’s students, tomorrow’s scientists.





WEEK 1

Kickstarting their journeys in STEM



DAY

TOPIC/ACTIVITY

MATERIALS

1

Distribution of pre-test (10 mins)

Introduction to STEM (10 mins)

This is the start of your students' journey in STEM! Discuss the significance and benefits of STEM with them in this section, as well as how it has evolved over time. Following that, facilitate a small group discussion in which they can share their perspectives on STEM and the types of support they require in STEM. Here are some suggested questions to consider during your discussion, but feel free to add your own:

1. What interests you the most about learning?
2. What is your opinion of STEM?
3. Have you encountered biases and stereotypes while studying STEM?
4. Do you have any reservations or concerns?
5. How can we help you navigate and support the STEM strand more effectively?

Preview of Toolkit and Slambook (5 mins)

1. Discuss the features and contents of the materials. Encourage them to fill out the slambook sections! (About Me, etc.)
2. Remind the students that the toolkit is a PDF with interactive buttons and links.
3. The slambook can also be edited.

Notes:

- You may open using any PDF application, software, or through Google Chrome, as long as you can fill up or edit the file.
- If you're using Chrome, make sure to save your changes before closing the PDF (look for the download arrow next to the printer icon in the upper right corner, then click save or "With Your Changes").

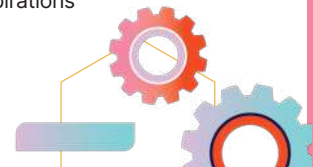
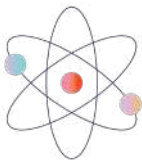
PCS Toolkit

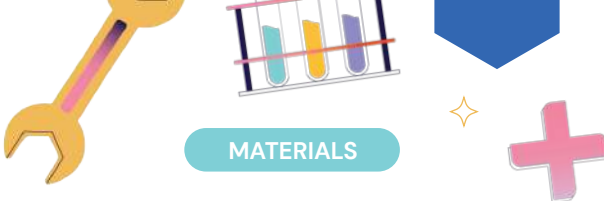
- **What is STEM?**
(pp. 4-5)
- **Best in Class: Preparing SHS students for STEM**
(pp. 10-13)

- **PCS Toolkit**
- **Slambook**

Discuss topics for each month

- Week 1: Intro to STEM
- Week 2: Role Models
- Week 3: STEM Experiments and Resources
- Week 4: STEM careers and aspirations





DAY

2

TOPIC/ACTIVITY

Strands, careers, and role models in STEM (10 mins)

Introduce students to the wonders and opportunities of STEM through stories, careers, and real-world applications. You can use the toolkit as a starting point for this discussion. Each section (Science, Technology, Engineering, and Mathematics) covers career and role model topics to assist students in visualizing their STEM paths. You may use your own examples or references to back up your points. Emphasize the importance of having more women in STEM fields. Here are some sample questions to ask them:

1. What or who inspired you to pursue a career in STEM?
2. What continues to inspire you to pursue a career in STEM?
3. Which STEM careers do you want to investigate?
4. Do you have a role model?

Class activity: My STEM Strengths (10 mins)

What STEM superpower do you have? List your strengths! It does not have to be grand; it can be something tiny that you know you can grow in!

After the activity, ask the students to share what they wrote in their slambook.

MATERIALS

PCS Toolkit

- **Exploring the world of science** (pp. 14-17)
- **Decoding the language of technology** (pp. 18-21)
- **Gearing up for engineering** (pp. 22-25)
- **Finding the x, y, and love for math** (pp. 26-29)

Slambook

- **My STEM Strengths** (p. 14)

✦ HOMEWORK ✦

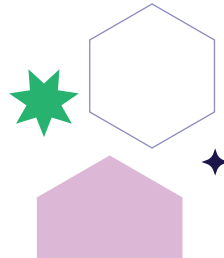
Power Quest #1: My STEM All Stars

1. Browse and read/watch the stories of Pinays in STEM.
2. Fill out the "My STEM Heroes" page from the slambook and list down five (5) stories that have influenced them the most.
3. Prepare for a short discussion in the next session.

PCS Toolkit

- **Stories of Pinays in STEM** (pp. 34-65)

Slambook

- **My STEM All Stars** (pp. 20-22)
- 



WEEK 2

Introducing role models in STEM



DAY

1

TOPIC/ACTIVITY

HW Discussion: My STEM All Stars (10 mins)

Use the slambook activity to lead a group discussion with your students. Encourage them to share their work!

If the students are comfortable, they may write a short social media post or blog online about their role model. Remind them to tag Pinays Can STEM (on Facebook and Instagram) so we can share! Students who share will receive Pinays Can STEM freebies from us!

Finding your role model (5 mins)

Role models are critical to every girl's confidence and participation in STEM. Giving girls access to them and highlighting their personal narratives, achievements, and journeys is a powerful step in the right direction for eradicating gender stereotypes and getting them excited about their own STEM path.

In this session, you will discuss the stories of the role models featured in the toolkit. Highlight key learnings and notable anecdotes to inspire and motivate them. You may also show some videos from our PCS TV to entice them (access the YouTube link through the toolkit or [here](#)).

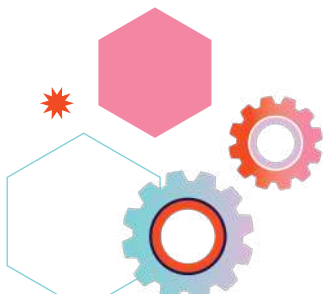
MATERIALS

Slambook

- **My STEM All Stars**
(pp. 20-22)

PCS Toolkit

- **Stories of Pinays in STEM**
(pp. 34-65)



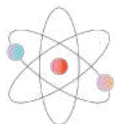
DAY

1

✦ HOMEWORK ✦



2



TOPIC/ACTIVITY

Power Quest #2: [Option 1] Get to know a role model in STEM

Encourage students to contact Pinays in STEM who they admire. Students, whether they are parents, teachers, or scientists, have the freedom to learn about their personal stories in order to stay motivated and inspired.

If they can meet a role model, they may take a "selfie" and post it online! Remind them to tag Pinays Can STEM so we can feature them, or they may email us through pinayscanstem@gmail.com.

Power Quest #2: [Option 2] Befriend a fellow Pinay classmate in STEM

Students can use the slambook to meet and get to know their classmates, as well as their own STEM goals.

Students may take a "selfie" with their new friend and post it online! Remind them to tag Pinays Can STEM so we can feature them, or they may email us through pinayscanstem@gmail.com. Selected Pinays will receive freebies from us.

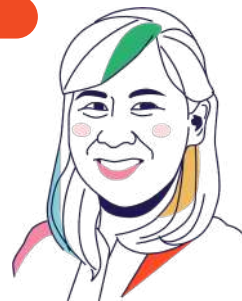
HW Discussion: Getting to know a role model/Friend-folio (5 mins)

Ask the students to share their homework from the previous week!

TIP: Check out our templates and posters to inspire your students to promote the cause! The link can be found in the toolkit. Make sure they tag Pinays Can STEM so we can reshare!



MATERIALS



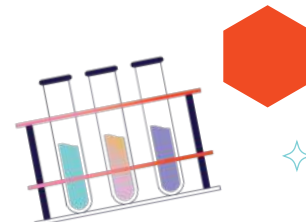
Slambook

- Friend-folio (pp. 5-9)

*Students can still use the Friend-folio page even if they don't choose this activity. They can also do both options.

PCS Toolkit

- Hype the advocacy (p. 77)





WEEK 3

STEM Experiments and Resources



DAY

TOPIC/ACTIVITY

MATERIALS

1

Let's find the fun in STEM! (10 mins)

Discuss with your students how STEM can be used outside of the classroom. This is significant because it demonstrates to students that STEM exists outside of the four walls of your classroom. It is critical, especially in this COVID-19 pandemic, to pique students' interest through hands-on activities. You could talk about the following:

1. Discuss the significance of STEM beyond what they learn in class.
2. Why should we study STEM? Why are STEM skills important in addition to STEM careers?
3. Access to STEM learning resources (websites, books, videos, etc.)

✦ **HOMEWORK** ✦**Power Quest #3: Exploring STEM resources**

STEM is all around us! Allow your students to listen to, watch, or read any of the toolkit materials. Students are strongly encouraged to engage with a variety of resources (1 podcast, 1 video) to broaden their STEM knowledge. You could make them listen to one of each.

Afterwards, students must complete the "Exploring the STEM Library" section of their slambook. Prepare to talk about it during the next session.

Note: Teachers may opt to dedicate time during Computer classes for students to navigate the STEM Library in school.

PCS Toolkit

- **PCS TV - My STEM Work videos** (pp. 62-65)
- **The STEM Library** (pp. 67-75)

PCS Toolkit

- **The STEM Library** (pp. 67-75)

Slambook

- **Exploring the STEM Library** (pp. 35-37)





DAY

2

TOPIC/ACTIVITY

MATERIALS

HW Discussion: Exploring STEM resources (10 mins)

Students will discuss their slambook entries and what they learned from the resources. The slambook can be used as a guide for the discussion.

PCS Toolkit

- **The STEM Library**
(pp. 67-75)

Slambook

- **Exploring the STEM Library**
(pp. 35-37)

Encouraging wonder in STEM (15 mins)

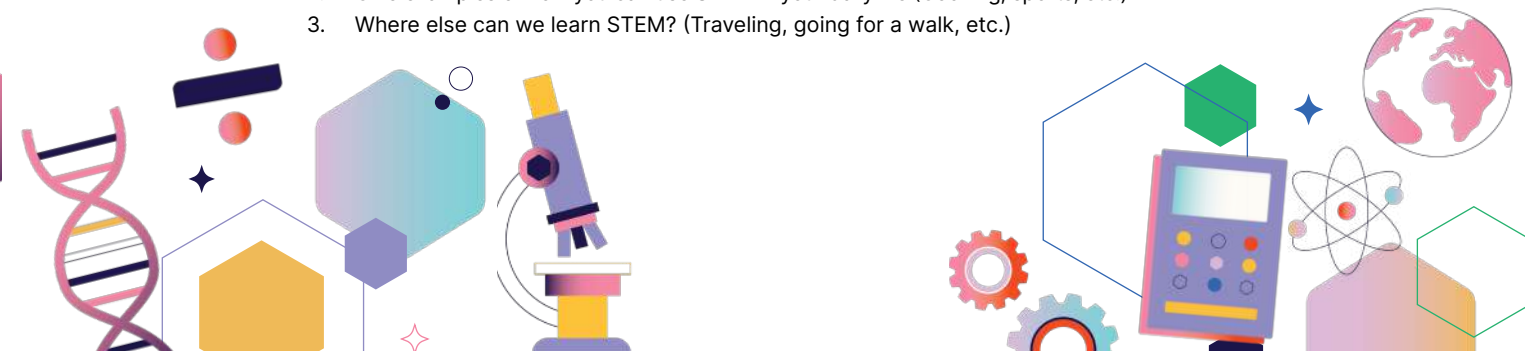
STEM is all around us! Allow your students to listen to, watch, or read any of the toolkit materials. Students are strongly encouraged to engage with a variety of resources (1 podcast, 1 video) to broaden their STEM knowledge. You could make them listen to one of each.

Materials

- 1 small sheet/strip of paper

Here are some guide questions for your students during your discussion:

1. Let students wonder. On any sheet of paper, finish the sentence "I've always wondered...." (For example, I've always wondered why water remains transparent no matter where it is placed.)
2. Give examples of how you can use STEM in your daily life (Cooking, sports, etc.)
3. Where else can we learn STEM? (Traveling, going for a walk, etc.)



DAY

2

TOPIC/ACTIVITY

Class activity: STEM experiments (20 mins)

Ask students to select 1-2 experiments from the toolkit that they want to try. In the slambook, students must document their observations and eureka moments after conducting STEM experiments in the slambook. Make sure they also list the STEM concepts that were used in the activity.

Encourage students to take pictures of their experiments, post them on a blog, or share them on social media. Please remind them to tag Pinays Can STEM so we can feature them or email us at pinayscanstem@gmail.com. Selected Pinays will receive freebies from us.

MATERIALS

PCS Toolkit

- **STEM Experiments** (p. 76)
- **Slambook**
- **STEM experiments journal** (pp. 25-34)

*Make sure to ask the students to prepare the materials beforehand

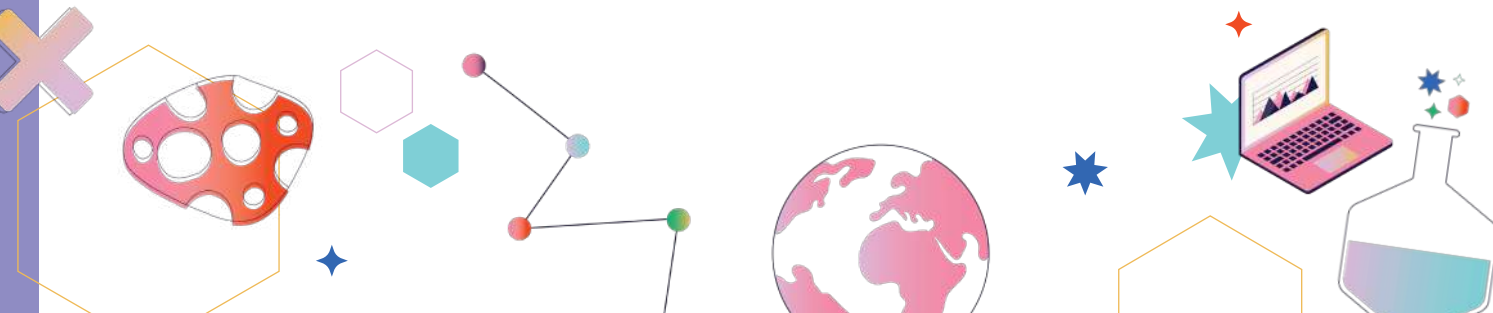
✦ HOMEWORK ✦

Power Quest #4: My STEM Milestones

Students should make a list of their STEM journey's goals and milestones. The plans are divided into three (3) sections: Big, Medium, and Small. Big milestones are their "big-picture" ambitions, medium goals are what they can achieve in the next three years (roughly during their university years), and small goals are what they can achieve daily. Next week, prepare for a group discussion.

Slambook

- **My STEM Milestones** (pp. 10-13)





WEEK 4

STEM careers and aspirations



DAY

1

TOPIC/ACTIVITY

HW Discussion: My STEM Milestones (5 mins)

Ask a few of your students to share their personal milestones. Here are some guide questions to get the conversation started:

1. Share a big, medium, or small goal.
2. What are your specific milestones under this goal?
3. Are there any obstacles to achieving your goals that you are willing to share?
4. What drives you to accomplish this goal?

MATERIALS

Slambook

- **My STEM Milestones**
(pp. 10-13)

✦ HOMEWORK ✦

Class activity: Journey to my career (5 mins)

For this session, explain the “Journey to my career” section in their slambook. Prepare for a discussion during the next session.

PCS Toolkit

Slambook

- **Journey to my career**
(pp. 15-16)



DAY

2

TOPIC/ACTIVITY

Discussion: Journey to my career (20 mins)

Assemble your students into groups and allow them to share in their circles. Following that, gather as a class and choose a few to share. There is no need to go through each question; instead, use the following guide questions to guide your discussion:

1. How do you see yourself in the future?
2. Choose one (1) of the items you've written down and tell the class about the career you want to pursue! You can explain why you want to take this path and who or what has inspired you.
3. What specific steps do you need to take to achieve this?
4. What do you hope to accomplish with your career? It can be internal (personal development, learning) or external (growth, learning) (to save the environment, treat diseases, fight misinformation, etc.)

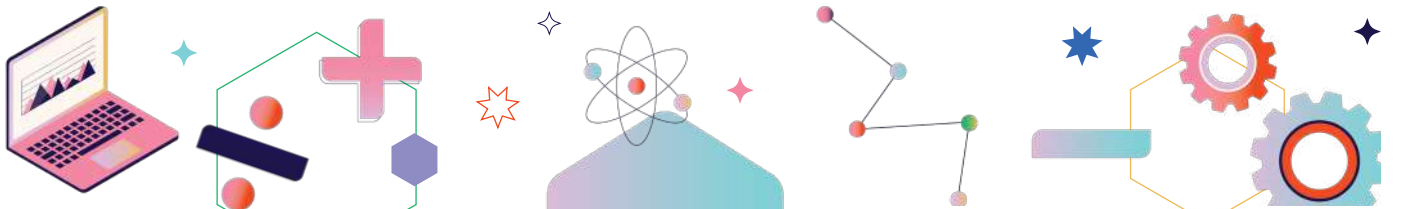
Class discussion:

1. Ask the students about their learnings and insights.
2. How do you feel now that you've completed the activity? How confident are you in your career now, on a scale of 1-10?
3. What type of assistance do you require from your parents, schools, and peers?
4. What was your main takeaway from this activity?

MATERIALS

Slambook

- **Journey to my career**
(pp. 15-16)



DAY

2

✦ HOMEWORK ✦

3

TOPIC/ACTIVITY

Power Quest #5: What I learned this month

Sharing of lessons and learnings from role models, resources, and other activities with the toolkit. Students can also share STEM lessons, concepts, and other fun trivias that they learned.

HW Discussion: What I learned this month (5 mins)

Allow students to share their toolkit learnings. Remember that students can continue to use the toolkit and slambook after the 1-month activity engagement period has ended.

RECAP (15 mins)

Give a brief summary of the previous month's activities. You may share some insights from the students' discussions, or you may use the contents of the toolkit as a reference. Here are some discussion points, though you may add more:

1. Emphasize the significance of STEM and role models.
2. Discuss the urgent need for more women in STEM, as well as the importance of diversity in the field.
3. Recap of Filipinas' achievements and contributions in STEM.
4. Gender stereotypes and misconceptions, and how to overcome them
5. Organizations to which they can reach out to.
6. If you're interested in taking part in the PCS community, please answer the evals form which will be distributed next week.

DISTRIBUTION OF POSTTEST, EVALS (15 mins)

MATERIALS

PCS Toolkit

Slambook

- **What I learned this month**
(pp. 23-24)

Slambook

- **What I learned this month**
(pp. 23-24)

PCS Toolkit

- **The STEM Directory**
(pp. 78-81)

Slambook





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